

What the Survey Missed

Teacher and presenter guide - bounded synthetic occupancy investigation - Murray State University - Bryant Harrison

Purpose and learning target

Students distinguish observed fraction ever detected, realized occupied-site fraction in one synthetic scenario, occupancy probability (ψ), and detection probability (p). Occupancy means whether a site is used in this model; it is not abundance or a number of animals.

Target claim: fewer detections do not by themselves establish lower occupancy. Detection probability can differ between habitats.

Every round visits the same 24 Habitat A and 24 Habitat B sites: 48 site-visits. A site click inspects a fixed detection history; it never changes the sample.

Current app flow

1. Open `index.html` locally; the **Guide** overlay opens automatically. Dismiss it, then run fixed round 1 across all 48 sites.
2. Inspect observed fraction ever detected and choose the cautious claim: a difference can reflect occupancy, detection probability, or both.
3. Only then choose fixed rounds 2 and 3, transfer, optional likelihood, or reveal.
4. After reveal, the same arrays distinguish detections, occupied-but-missed sites, and unoccupied sites. Further rounds are exploration, not a blind new test.

Controls: ? reopens the **Guide** at any point. **Settings** offers **Open Presenter Notes** (this guide, on screen), **Presentation mode** (hides the case aside, the transfer and reveal panels, and the likelihood view for a projector), and **Reset** (the whole demo back to first load: the first seeded scenario, no rounds run, no claims chosen, nothing revealed, the likelihood view closed and re-centred). Reset leaves Presentation mode as you set it. Two scenario controls sit below the rule: **Replay as practice (same seed)** keeps whatever scenario is on screen, and **New seeded scenario** draws the next one.

Declared synthetic model

Habitat	Generating occupancy ψ	Detection p	Meaning
A	0.8	0.25	Higher generating occupancy probability, lower detection probability.
B	0.5	0.8	Lower generating occupancy probability, higher detection probability.

Sites remain occupied or unoccupied across the three visits (closure). Given occupancy, detections are conditionally independent; there are no false positives; and p is constant within a habitat. These are teaching assumptions, not claims about local wildlife.

Why raw detections can mislead

For K visits, expected fraction ever detected is $\psi \times [1 - (1 - p)^K]$.

Visits K	A expected ever detected	B expected ever detected	Implication
1	0.20	0.40	B is expected to have more detections although A has greater psi.
3	0.4625	0.496	More visits improve evidence but do not guarantee a finite ranking matches occupancy.

Do not score the ranking. After reveal, compare observed fraction, realized occupancy, generating psi, and p by name. A seeded realization can differ from either expectation.

Optional likelihood view

The app uses observed K-visit histories for one habitat only. For an ordered history with d detections, likelihood is $\psi^d (1-\psi)^{K-d}$ when d is positive; for an all-zero history it is $(1-\psi) + \psi(1-\psi)^K$.

Interpret carefully: shading is relative likelihood on a bounded psi/p grid. It is not posterior probability or a calibrated confidence interval. Multiply exact ordered-history probabilities over sites; do not add a binomial coefficient to one particular history.

With one visit, detections chiefly identify psi p. A high-psi/low-p and lower-psi/higher-p combination can fit the same observed fraction. The app supplies axis labels plus keyboard/touch readout rather than a unique occupancy estimate.

Facilitation and boundaries

- Before round 1, contrast “not detected” with “not present.”
- After round 1, ask for the unit of effort: 48 site-visits, not 48 animals counted.
- Before another round, ask what changes (histories) and what stays fixed (sites).
- For transfer, ask whether a different detection process could explain fewer detections.

This activity teaches detection confounding. It does not estimate a real species, habitat, population, conservation priority, or field-survey result.

Sources and attribution

MacKenzie, D. I., et al. (2002), “Estimating Site Occupancy Rates When Detection Probabilities Are Less Than One,” *Ecology*, 83(8), 2248-2255. U.S. Geological Survey: [Estimating site occupancy rates when detection probabilities are less than one](#).

U.S. Geological Survey, [RPresence occupancy workshop, 2023 online](#), for occupancy/detection-history context. These are source context, not course assignment information or endorsement.

What the Survey Missed - Bryant Harrison - Murray State University - Offline synthetic teaching model. This HTML file is the canonical guide source: the printable sheet, the PDF, and the in-app presenter notes all come from it.